

ASTRAOS

Professional Open-Source Operating System Project Proposal

Executive Summary

AstraOS is a modern educational operating system built completely from scratch. It aims to give contributors practical experience in low-level programming, computer architecture, memory management, multitasking, file systems, device drivers and operating system design while following professional software engineering practices.

Vision

To build a clean, modular, open-source operating system that becomes a learning platform for students and showcases real engineering skills.

Mission

Create an extensible OS architecture with maintainable code, excellent documentation, GitHub collaboration, code reviews and continuous integration.

Core Features

- Custom Bootloader
- 64-bit Kernel
- Memory Management & Paging
- Interrupt Handling (GDT/IDT)
- Process Scheduler
- Multitasking
- AstraFS Custom File System
- Keyboard & Mouse Drivers
- System Calls
- Interactive Shell
- Basic GUI
- Basic Networking
- Testing & Documentation

Technology Stack

C, C++, x86-64 Assembly, NASM, GCC Cross Compiler, Make, QEMU, GDB, Git, GitHub Actions.

Development Methodology

Agile development with weekly milestones, pull requests, code reviews, issue tracking, branch protection, testing and documentation.

Repository Structure

astra-boot/
astra-kernel/
astra-memory/
astra-fs/
astra-drivers/
astra-shell/
astra-gui/
astra-network/
docs/
website/

Expected Deliverables

- Bootable ISO
- Public GitHub Repository
- Technical Documentation
- Architecture Diagram
- API Documentation
- Demo Video
- Final Presentation

Future Roadmap

Version 1.0 – Educational OS
Version 2.0 – GUI Improvements
Version 3.0 – Networking Stack
Version 4.0 – Package Manager
Version 5.0 – Security Enhancements

90-Day Roadmap

Phase	Duration	Modules
Phase 1	Days 1–30	Bootloader, Kernel, Memory, Interrupts
Phase 2	Days 31–60	Scheduler, Paging, AstraFS, Drivers
Phase 3	Days 61–90	GUI, Networking, Testing, Documentation, Release

Team Structure

Samir Levate – Team Leader & Founder: Architecture, Project Planning, Integration, GitHub, Documentation.

Shubham Gaikwad – Co-Founder & HR: Recruitment, Team Coordination, Progress Tracking.

Additional Contributors: Kernel Developer • Memory Engineer • File System Engineer • Driver & GUI Engineer • QA & Documentation.

Why Join AstraOS?

Contributors gain real-world systems programming experience, open-source collaboration, GitHub contributions, portfolio-quality work, code review experience, and exposure to professional development workflows.